

ELECTRICAL FUNDAMENTALS COMPETENCY



Industry Training Authority of BC
Camosun College

Book: Electrical Fundamentals

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Foreword

The BC Open Textbook Project began in 2012 with the goal of making post-secondary education in British Columbia more accessible by reducing student cost through the use of openly licensed textbooks. The BC Open Textbook Project is administered by BCCampus and is funded by the British Columbia Ministry of Advanced Education.

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Preface

The concept of identifying and creating resources for skills that are common to many trades has a long history in the Province of British Columbia. This collection of Trades Access Common Core (TACC) resources was adapted from the 15 Trades Common Core line modules co-published by the Industry Training and Apprenticeship Commission (ITAC) and the Centre for Curriculum Transfer and Technology (C2T2) in 2000-2002. Those modules were revisions of the original Common Core portion of the TRAC modules prepared by the Province of British Columbia Ministry of Post-Secondary Education in 1986. The TACC resources are still in use by a number of trades programs today and, with the permission from the Industry Training Authority (ITA), have been utilized in this project.

These open resources have been updated and realigned to match many of the line and competency titles found in the Province of BC's trades apprenticeship program outlines. A review was carried out to analyze the provincial program outlines of a number of trades, with the intent of finding common entry-level learning tasks that could be assembled into this package. This analysis provided the template for the outline used to update the existing modules. Many images found in ITA apprentice training modules were also incorporated into these resources to create books that are similar to what students will see when they continue their chosen trades training. The project team has also taken many new photographs for this project, which are available for use in other trades training resources.

The following list of lines and competencies was generated with the goal of creating an entry-level trades training resource, while still offering the flexibility for lines to be used as stand-alone books. This flexibility—in addition to the textbook content being openly licensed—allows these resources to be used within other contexts as well. For example, instructors or institutions may incorporate these resources into foundation-level trades training programming or within an online learning management system (LMS).

Line A – Safe Work Practices

- A-1 Control Workplace Hazards
- A-2 Describe WorkSafeBC Regulations
- A-3 Handle Hazardous Materials Safely
- A-4 Describe Personal Safety Practices
- A-5 Describe Fire Safety

Line B – Employability Skills

- B-1 Apply Study and Learning Skills
- B-2 Describe Expectations and Responsibilities of Employers and Employees
- B-3 Use Interpersonal Communication Skills
- B-4 Describe the Apprenticeship System

Line C – Tools and Equipment

- C-1 Describe Common Hand Tools and Their Uses

- C-2 Describe Common Power Tools and Their Uses
- C-3 Describe Rigging and Hoisting Equipment
- C-4 Describe Ladders and Platforms

Line D – Organizational Skills

- D-1 Solve Trades Mathematical Problems
- D-2 Apply Science Concepts to Trades Applications
- D-3 Read Drawings and Specifications
- D-4 Use Codes, Regulations, and Standards
- D-5 Use Manufacturer and Supplier Documentation
- D-6 Plan Projects

Line E – Electrical Fundamentals

- E-1 Describe the Basic Principles of Electricity
- E-2 Identify Common Circuit Components and Their Symbols
- E-3 Explain Wiring Connections
- E-4 Use Multimeters

All of the self-test questions are also available from BCcampus as separate data, if instructors would like to use the questions for online quizzes or competency testing.

Industry Training Authority of BC

The ITA works with employers, employees, industry, labour, training providers, and government to issue credentials, manage apprenticeships, set program standards, and increase opportunities in approximately 100 BC trades. Among its many functions are oversight of the development of training resources that align with program standards, outlines, and learning objectives, and authorizing permission to utilize these resources (text and images).

- Erin Johnston, Director of Training Delivery
- Cory Williams, Manager, Industry Relations
- Publishing Services, Queen's Printer
- Spencer Tickner, Director of QP Publishing Services
- Dwayne Gordon, Manager, Electronic Publishing

SECTION OVERVIEW

1: Unit I - Basic Principles of Electricity

Learning Objectives

When you have completed the Learning Tasks in this Competency, you will be able to:

- describe the composition of matter and the structure of the atom
- describe the principles of electricity and the theory of current flow
- describe basic types of electrical circuits and their characteristics
- describe electromagnetism

You will use electricity daily. Vehicles and machinery are started and often operated by electricity. Electric tools make the performance of your job easier and more efficient. However, to use electricity safely and effectively, it is important to understand electricity's terminology and principles.

1: Fundamentals of Electricity

1.1: Basic principles

1.2: Electrical circuits and units of measurement

1.E: Self-Test 1

2: Basic Circuit Concepts

2.1: Basic Electrical Circuits

2.2: Series Circuits

2.3: Parallel Circuits

2.4: Series-parallel Circuits

2.5: Polarity and direction of current flow

2.E: Self-Test 2

3: Electromagnetism

3.1: Magnetic Fields

3.2: Electricity and Magnetism

3.E: Self-Test 3

4: Answer Key

Thumbnail: Interior Communications Electrician 3rd Class Karla Martinez inspects wiring aboard the aircraft carrier USS Nimitz. (Public Domain; US Navy via [Wikipedia](#))

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CHAPTER OVERVIEW

1: Fundamentals of Electricity

Over the centuries scientists have discovered that electricity is predictable and measurable. Being familiar with the fundamentals of electricity will help you to understand how and why electrical circuits work.

[1.1: Basic principles](#)

[1.2: Electrical circuits and units of measurement](#)

[1.E: Self-Test 1](#)

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1.1: Basic principles

Learning Task 1

Explain fundamentals of electricity

Over the centuries scientists have discovered that electricity is predictable and measurable. Being familiar with the fundamentals of electricity will help you to understand how and why electrical circuits work.

Basic principles

Electricity is a form of energy. To understand electricity, it is important that you first understand the structure of matter. Anything that occupies space and has weight is called matter. All liquids, gases, and solids are examples of matter in different forms. Matter is made of smaller units called atoms. Atoms can be grouped together in compounds to form molecules.

Atomic theory

Atoms are the most basic part of matter and differ in atomic structure from each other. The structure of the atom can be described in much the same way as the solar system. Instead of the Sun at the centre, there is a nucleus. This nucleus is made of two basic particles: protons and neutrons.

Neutrons make up the mass (or weight equivalency) of the atom, have no electrical charge, and are considered to be neutral. Protons are particles that have a positive (+) electrical charge and cannot be separated from the nucleus. Surrounding the nucleus in orbits are electrons. These are tiny particles with a negative (–) electrical charge. Figure 1 shows a model of a carbon atom.

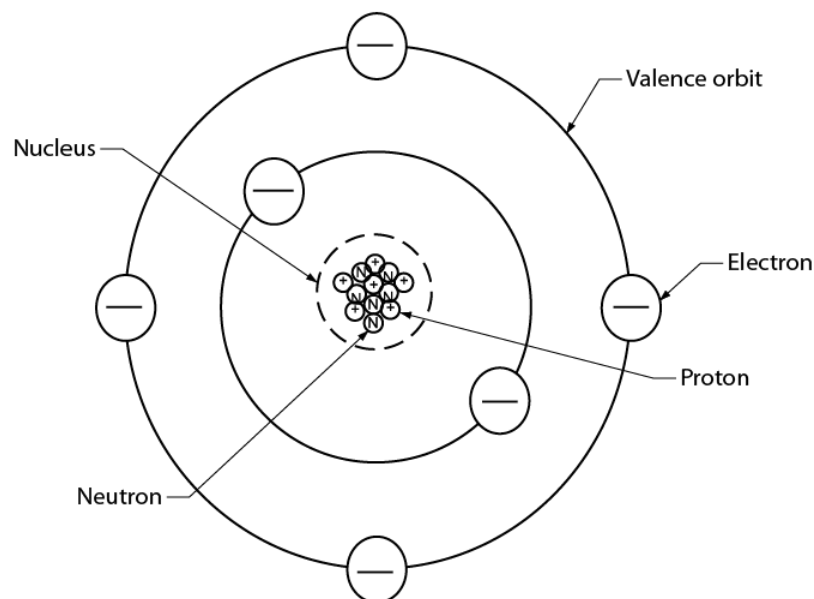


Figure 1.1.1: (CC BY-NC-SA; BC Industry Training Authority)

1. Carbon atom

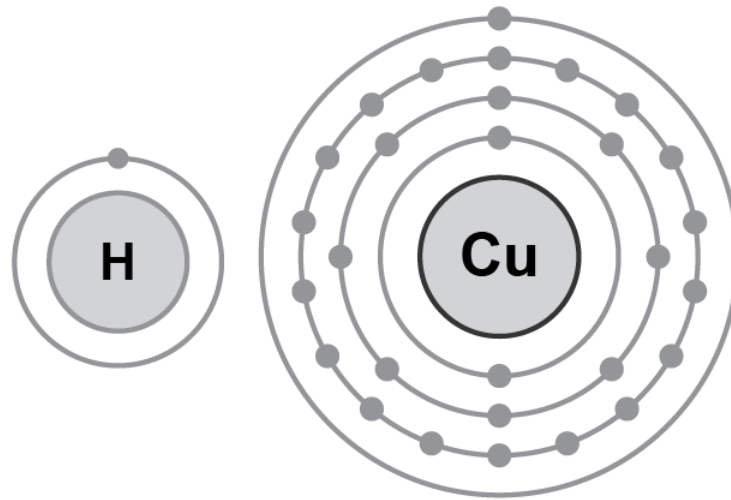


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2. Hydrogen and copper atoms

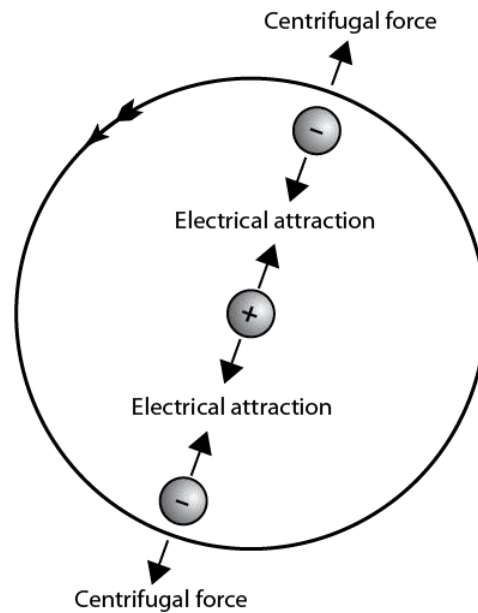


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3. Electrical attraction

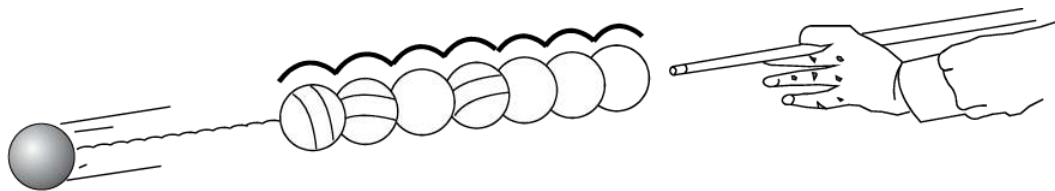


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4. Transmission of impulse

Sources of electrical force

You have just learned that if there is a surplus of electrons at one end of a conductor and a deficiency at the other end, a current flows in the conductor. There are devices that create this difference in charge so that a current will flow. These devices are referred to as sources of electromotive force. These sources include:

- chemical

- electromagnetic induction
- friction
- heat
- pressure
- light

Chemical

A battery is a source of electrical force due to the chemical reaction that takes place between plates and an electrolyte. This reaction causes a buildup of positive ions on one plate and negative ions on the other plate. This electrical difference between the plates is also known as potential difference.

Electromagnetic induction

Electric force can be generated by using a magnetic field. This is the method by which most of the electrical energy we use is produced. An example is an alternator or generator.

Friction

Friction can cause free electrons to move from one body to another and be stored there temporarily. When you walk across a carpet, electrons are transferred to the atoms in your body and you return them to other atoms when you touch a metallic object.

Heat

If two unlike metals are placed together and heated, they will produce electrical force. An example is the thermocouple in a furnace.

Pressure

Certain crystals will produce electricity if they are squeezed under extreme pressure. An example is a barbecue starter (also called piezoelectric generator).

Light

Some crystals and semiconductors will produce electrical force when they are exposed to light. An example is the photocell in a calculator.

All six of these sources of EMF achieve the same thing. They separate charge by:

- imparting energy to the electrons
- pushing them against an electrostatic field
- causing a surplus of electrons (negative charge) at one terminal of the source and a deficiency of electrons (positive charge) at the other terminal

In a sense, the process can be likened to compressing a spring. The energy stored in the compressed spring can be used later to do useful work. The same is true of the separated charges: they store energy that can be used to do useful work.

Electrical energy always comes from some other form of energy. The source of EMF is simply the device that makes the conversion from some other form of energy to electrical energy.

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CHAPTER OVERVIEW

3: Electromagnetism

A magnet attracts ferrous metals and some alloys. Magnets can take three forms:

- natural
- artificial
- electric

Natural magnets (i.e., magnetite) are very weak. Artificial magnets are made from magnetic materials (such as iron, nickel, and cobalt) and are given a strong magnetic force during construction. These are permanent magnets and have some limited use. Electromagnets can be easily turned on and off and are in common use because they are not permanent. They are called temporary magnets.

[3.1: Magnetic Fields](#)

[3.2: Electricity and Magnetism](#)

[3.E: Self-Test 3](#)

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3.2: Electricity and Magnetism

Electricity and magnetism

There is a direct relationship between electricity and magnetism. If there is current flow in a conductor there will be lines of force created around the conductor. If you could look at the magnetic field formed around a current-carrying conductor, it would look like Figure 3.2.1.

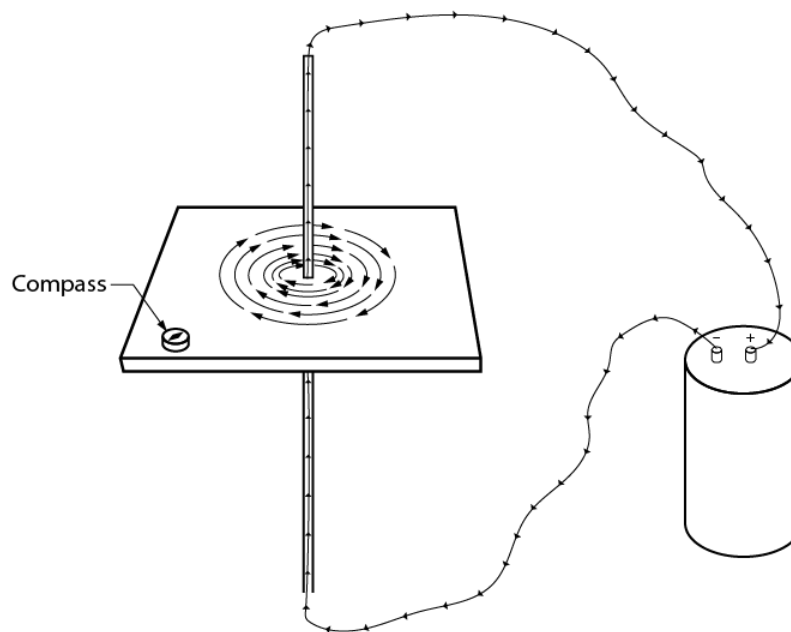


Figure 3.2.1: Electromagnetic field (CC BY-NC-SA; BC Industry Training Authority)

Note that the lines of force circle the conductor in rings and have direction. The direction of the lines of force depends on the direction of electron flow. If you know the direction of electron flow, you can determine the direction of the lines of force by using your left hand.

The “left-hand rule” says that if you hold the conductor in your left hand with your thumb pointing in the direction of the electron current flow, your fingers will curl in the direction of the lines of force. You will sometimes find this referenced as the “right-hand rule” from those using convention flow notation.

Interaction of fields

Magnetic fields around a current-carrying conductor act in the same way as the fields around a permanent magnet. In Figure 3.2.2, two conductors have been moved close together. The current is going in opposite directions, as indicated by the symbol in the end of the conductor. An X indicates electron flow in; a dot indicates electron flow out. The magnetic lines of force try to push the two conductors apart because they are in opposite directions. The arrows indicate the direction of the magnetic force.

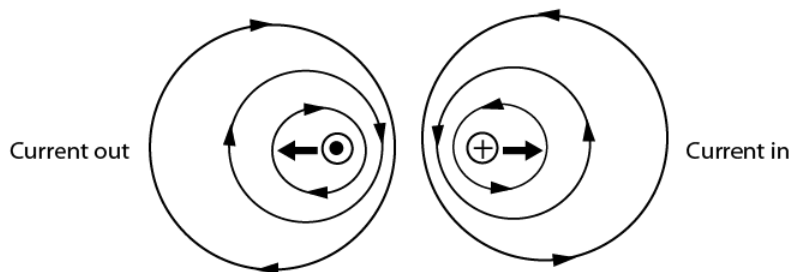


Figure 3.2.2: Opposite currents repel (CC BY-NC-SA; BC Industry Training Authority)

If one of the conductors has the current reversed, then the magnetic lines of force travel in the same directions. When this occurs the lines of force try to contract and pull tight, just as they did with a permanent magnet. The resulting force will try to pull the two

conductors together (Figure 3.2.3).

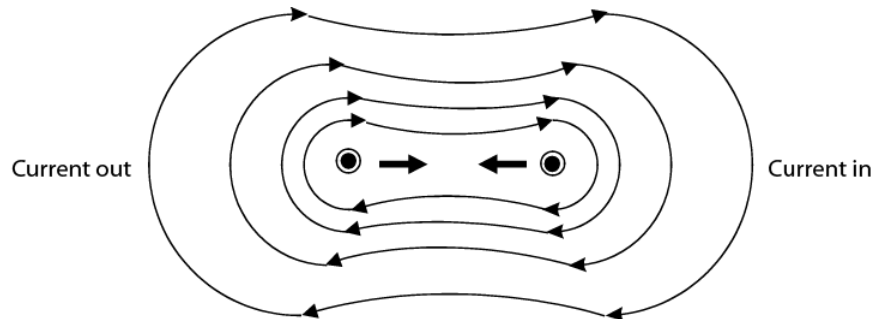


Figure 3.2.3: Like currents attract (CC BY-NC-SA; BC Industry Training Authority)

Conductors in loops

If a conductor carrying a current is formed into a loop, the magnetic field will be arranged differently. It will form looped lines of force with a north pole on one side of the loop and a south pole on the other. The magnetic flux lines add to each other and produce a much denser magnetic field in the center of the coil (Figure 3.2.4).

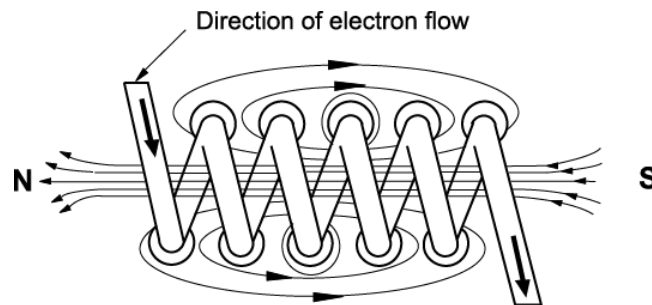


Figure 3.2.4: Flux in a loop (CC BY-NC-SA; BC Industry Training Authority)

Electromagnets

If a piece of soft iron is placed in the coil and a current is passed through the coil in one direction, the magnetic field of the coil causes the domains to align in the iron. This causes poles to form in the iron and creates an electromagnet, as shown in 3.2.5.

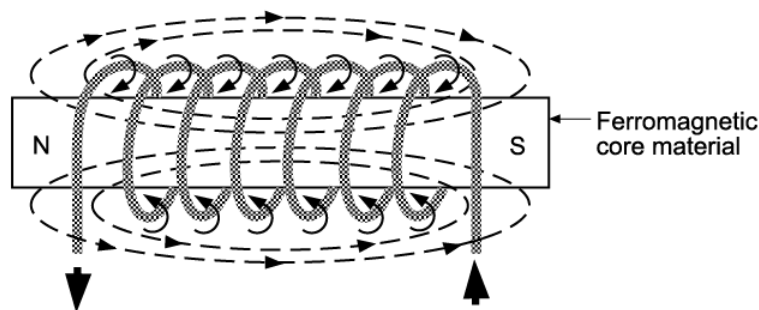


Figure 3.2.5: Electromagnet (CC BY-NC-SA; BC Industry Training Authority)

The strength of the electromagnet varies with the number of loops formed, the strength of the electric current, and the type of core in the winding. Because the iron core has a low magnetic retention, the magnetic field collapses when the current stops flowing. The iron core is no longer magnetized and will release whatever it was being used to hold or pull inward.

Electric motors and generators

Electromagnets are probably most commonly used in motors and generators. We have seen that magnetism can be caused by electricity. Electric motors use the force of electromagnets to produce rotation. On the other hand, electricity can be produced by magnetism. When a conductor is moved through a magnetic field or a magnet is moved past a conductor, the movement will induce a voltage in the conductor. Most electricity is generated in this way.

To generate a voltage, three elements must be combined (Figure 3.2.6:

1. a conductor
2. a magnetic field
3. movement by either the conductor or the magnetic field

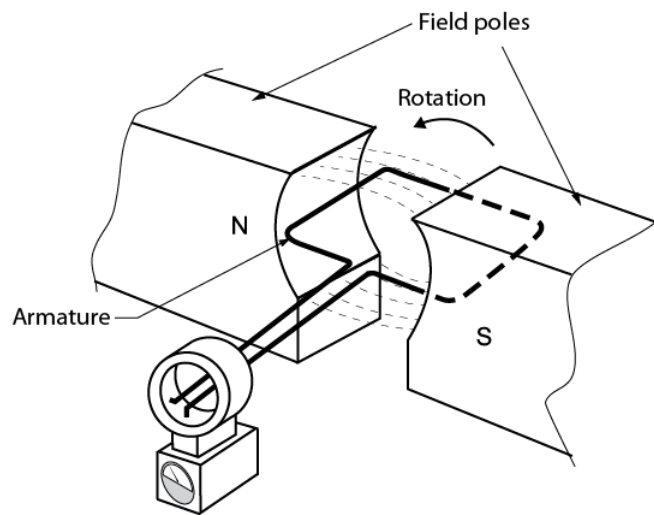


Figure 3.2.6: Simple generator (CC BY-NC-SA; BC Industry Training Authority)

The amount of voltage produced will depend on the strength of the magnetic field and the speed at which the conductor or the field moves. A conductor that moves through a magnetic field quickly will generate a higher voltage than one that moves more slowly.

A conductor that moves through a strong magnetic field will generate a higher voltage than one moved through a weak magnetic field.

Alternating current

Electric current that flows in one direction for a split second then changes direction in another split second is called alternating current (AC). In an alternating voltage, the polarity reverses direction periodically. The spinning mechanical motion of an electric generator produces AC voltage and current.

AC waveform and hertz

Hertz is the unit used to describe the frequency of AC direction change. Figure 3.2.7 is a graphic illustration using a curved line with arrows to indicate a change of direction in AC electron flow. Starting at point A, the current flows in one direction, and then at 120 volts it changes direction, drops to 0 volts, and continues to 120 volts, where it changes direction again back to point B at 0 volts.

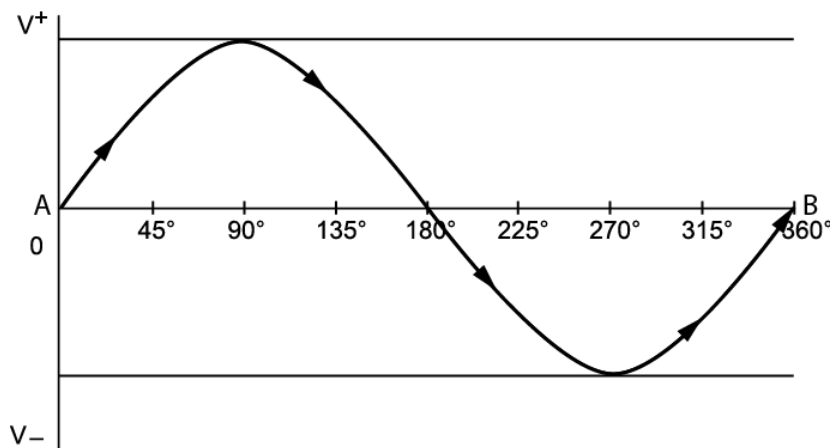


Figure 3.2.7: AC sine wave (CC BY-NC-SA; BC Industry Training Authority)

If it takes one second to complete the cycle from A to B, we would say the frequency is 1 hertz. Household utility AC current is supplied to the customer at 60 hertz, meaning 60 cycles per second.

Single-phase power supply

Single-phase electric power refers to the distribution of alternating current electric power using a system in which all the voltages of the supply vary in unison. Single-phase distribution is used when loads are mostly lighting and heating, and with a few large electric motors. Single-phase power typically comprises one voltage that is carried between two separate conductors. The two hot lines are called Line 1 and Line 2. In some systems, a grounded neutral, often labeled N, is provided that reduces the referenced voltage in half. These systems are typically found in residential and small commercial applications.

Three-phase AC power supply

Three-phase electrical power refers to a type of electrical power distribution in which three or more energized electrical conductors are carrying alternating currents. Examples of three-phase power systems are industrial applications and power transmission. Three-phase power supplies are used to power large motors and other heavy loads. A three-phase system is generally more economical than equivalent single-phase or two-phase (an uncommon power supply) systems at the same voltage.

Three-phase power comprises three independent voltages that are carried on three separate conductors. The three hot lines are called Line 1, Line 2, and Line 3. Three-phase power is typically found in commercial and industrial buildings.



Now complete the Learning Task Self-Test.

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SECTION OVERVIEW

2: Unit II- Common Circuit Components and Their Symbols

Learning Objectives

- When you have completed the Learning Tasks in this Competency you should be able to:
- describe the components of a circuit
- identify common electrical components
- describe solid state devices
- identify wiring connectors
- describe the composition of simple wiring diagrams

Many types of electric devices are used in almost every trade. Mechanical trades will find electrical devices in every motor-driven vehicle. Construction trades will encounter them in most construction projects. To be able to recognize or troubleshoot electrical devices, you must be familiar with the correct terminology and symbols used to identify them.

5: Circuit Components and their Schematic Symbols

5.1: Electrical systems

5.2: Conductors

5.3: Receptacles

5.4: Circuit Protection Devices

5.5: Circuit Control Devices

5.6: Lamps

5.7: Transformers

5.8: Solid state components

5.9: Capacitors

5.10: Light-emitting diodes (LEDs)

5.11: Sealed Components

5.12: Cells and Batteries

5.E: Self Test 1

6: Wiring Diagrams

6.1: Symbols Used in Schematic Diagrams

6.2: Types of Electrical Diagrams

6.E: Self Test 2

7: Common Circuit Characteristics

7.1: Series Circuits

7.2: Parallel Circuits

7.3: Voltage Source Circuits

7.4: Three-wire Power Supply System

7.E: Self-Test 3

8: Answer Key

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5.12: Cells and Batteries

Cells and batteries convert chemical energy to electrical energy. They are used to provide power to portable electronic equipment. They may also be used for backup power in the event of a power failure.

The single cells shown in Figure 24 can produce 1 to 4 volts, depending on their compositions. For example, cells may be carbon-zinc, alkaline-manganese, nickel-cadmium, silver oxide, or lithium types.



Figure 5.12.1: Cells (CC BY-NC-SA; BC Industry Training Authority)

Note the polarity of the symbol for a cell shown in Figure Figure 5.12.2.

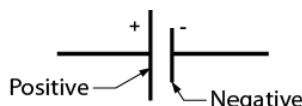


Figure 5.12.2: Symbol for a cell (CC BY-NC-SA; BC Industry Training Authority)

Cells may be connected in series within a case to make up a battery. Both the 12-volt automotive battery and the 9-volt clip-in battery (Figure 5.12.3) have multiple cells inside the case. The symbol for a battery is composed of the number of symbols for cells corresponding to the number of cells in the battery (Figure 5.12.4).



Figure 5.12.3: Batteries (CC BY-NC-SA; BC Industry Training Authority)

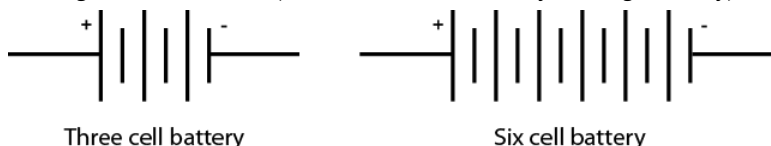


Figure 5.12.4: Symbols for batteries (CC BY-NC-SA; BC Industry Training Authority)



Now complete the Learning Task Self-Test.

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6.1: Symbols Used in Schematic Diagrams

Electrical symbols may be different for each manufacturer, but in most cases they are standard. Every manufacturer's diagram should have a symbol identification chart or "key" located in the wiring book. Some examples of electrical symbols are shown in Figure \(\PageIndex{1}\).

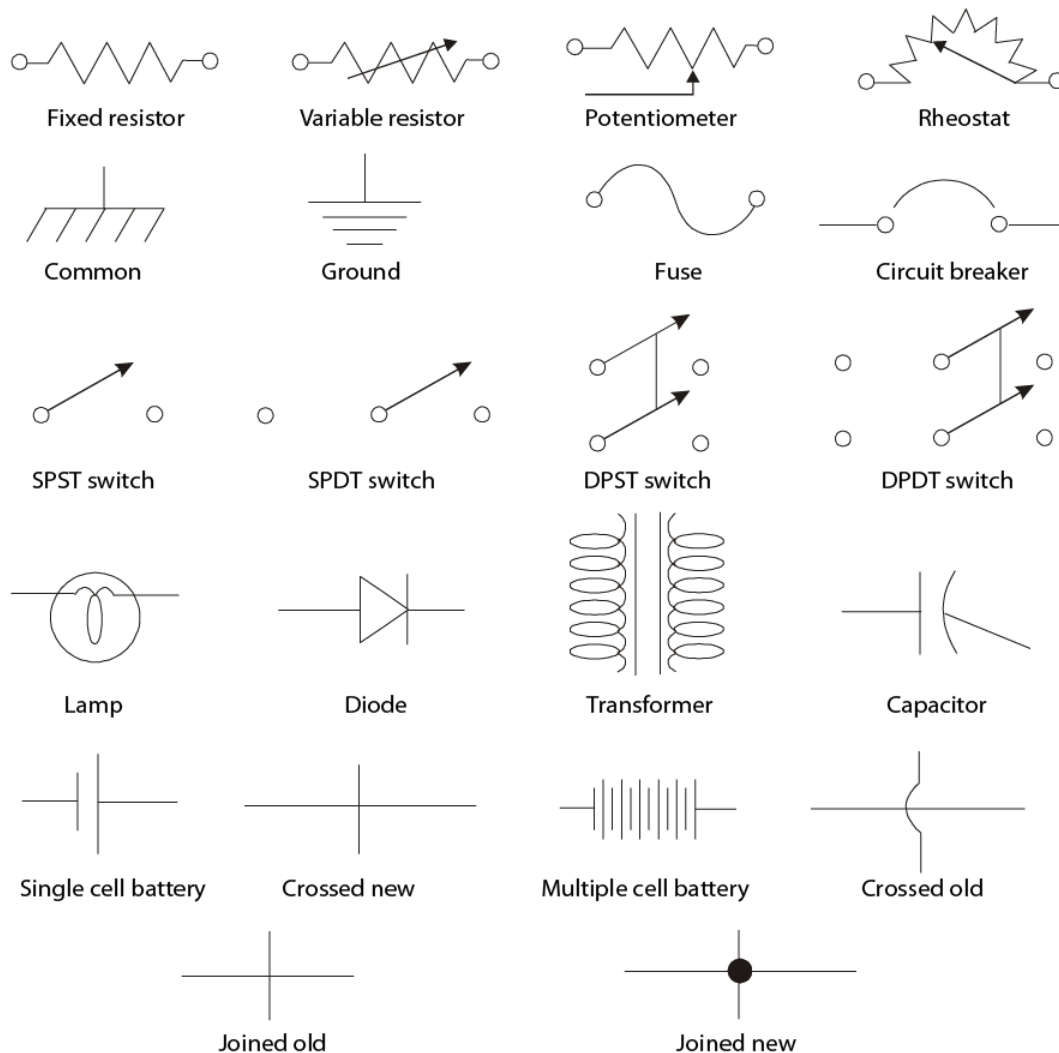


Figure 6.1.1: Common electrical symbols (CC BY-NC-SA; BC Industry Training Authority)

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6.E: Self Test 2

Self-Test 2

1. A schematic diagram is often called a ladder diagram.
 1. True
 2. False
2. What is the fourth type of wiring diagram, in addition to schematic, wiring, and block diagrams?
 1. Line
 2. Oblique
 3. Pictorial
 4. Isometric
3. A schematic diagram shows all physical locations of components.
 1. True
 2. False
4. For which operations is a wiring diagram best suited?
 1. Diagnosing
 2. Sequencing
 3. Installation
 4. Troubleshooting
5. A pictorial diagram is used to locate components in complex systems.
 1. True
 2. False
6. A block diagram includes symbols.
 1. True
 2. False

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7: Common Circuit Characteristics

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10.E: Self-Test 2

Self-Test 2

1. Safety factors such as toxic fumes and splashing flux should always be addressed when soldering.
 1. True
 2. False
2. Which of the following best describes the process of surface penetration?
 1. Bonding
 2. Wetting
 3. Soldering
 4. Prepping
3. Which of the following should apply when tinning stranded wire?
 1. The strands should be thickly coated.
 2. The strands should be visible under the solder.
 3. The solder should wick up underneath the insulation.
 4. A large blob of solder should form at the end of the wire.
4. Metals like aluminum and iron wet easily.
 1. True
 2. False
5. What is the purpose of flux?
 1. To remove oxides
 2. To clean the metal
 3. To assist with heating
 4. To remove the solder
6. What is the most common alloy used for solder?
 1. Tin-lead
 2. Lead-silver
 3. Tin-copper
 4. Lead-copper
7. Soldering pencils are used for heavy-duty applications.
 1. True
 2. False
8. What must be done to a soldering tip prior to soldering?
 1. Tin it.
 2. Wash it.
 3. Wire brush it.
 4. Heat it red hot.
9. What should be done to an overly corroded soldering tip?
 1. Nothing.
 2. Re-tin it.
 3. Throw it away.
 4. Dress it with a fine file.
10. What is the purpose of a heatsink?
 1. To protect the solder
 2. To pull flux to the joint
 3. To pull heat to the area for soldering

4. To pull heat away from certain components
11. What would cause a solder joint to be piled up and lumpy in appearance?
 1. Insufficient heat
 2. Lack of flux
 3. Poor tinning
 4. A corroded tip

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11: Answer Key

Answer Key

Self-Test 1

1. c. Conductor
2. d. Multiple insulated wires grouped together in a common sheathing
3. c. To prevent unwanted current flow
4. d. When flexibility is needed
5. b. False
6. b. It should be tinned.
7. b. False
8. d. It can be removed easily.
9. b. False
10. c. Crimp-on

Self-Test 2

1. a. True
2. b. Wetting
3. b. The strands should be visible under the solder.
4. b. False
5. a. To remove oxides
6. a. Tin-lead
7. b. False
8. a. Tin it.
9. d. Dress it with a fine file.
10. d. To pull heat away from certain components
11. a. Insufficient heat

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SECTION OVERVIEW

4: Unit IV- Using Multimeters

Learning Objectives

When you have completed the Learning Tasks in this Competency, you will be able to:

- describe the proper handling, use, and storage of electric meters
- describe the procedures for connecting meters to measure voltage, current, and resistance
- describe how to use meters to analyze electric circuits

This Competency will introduce you to three basic meters for measuring voltage, current, and resistance. You must have a basic understanding of the purpose and operation of each type of meter before you attempt to use one. If you connect a meter incorrectly, you not only risk damaging the instrument, but more importantly, you or some innocent bystander could receive a serious electrical shock.

12: Describe the use and storage of meters

12.1: Prelude to Describe the use and storage of meters

12.2: Meter safety precautions

12.3: Digital Multimeters

12.4: Introduction to voltage measurements

12.5: Introduction to current measurements

12.6: Introduction to resistance measurements

12.E: Self-Test 1

13: Use meters to analyze simple circuits

13.1: Use meters to analyze simple circuits

13.2: Troubleshooting principles

13.3: Polarity in a parallel circuit

13.E: Self Test 2

14: Answer Key

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13.E: Self Test 2

Self-Test 2

1. What is the lowest amperage at which fatal injury can occur in a human?
 1. 15 A
 2. 0.15 A
 3. 50 mA
 4. 15 mA
2. If you measure voltage across a switch, what state is it in?
 1. Open
 2. Closed
3. If you measure correct voltage across a load but the load doesn't work, what is wrong?
 1. There is no power.
 2. There is no current.
 3. A switch is open.
 4. The load has failed.
4. Refer to Figure 1. If you have a reading at point 4, but not at 5, what is the problem?
 1. The load has failed.
 2. The switch is open.
 3. The wiring has failed.
 4. The ground has failed.

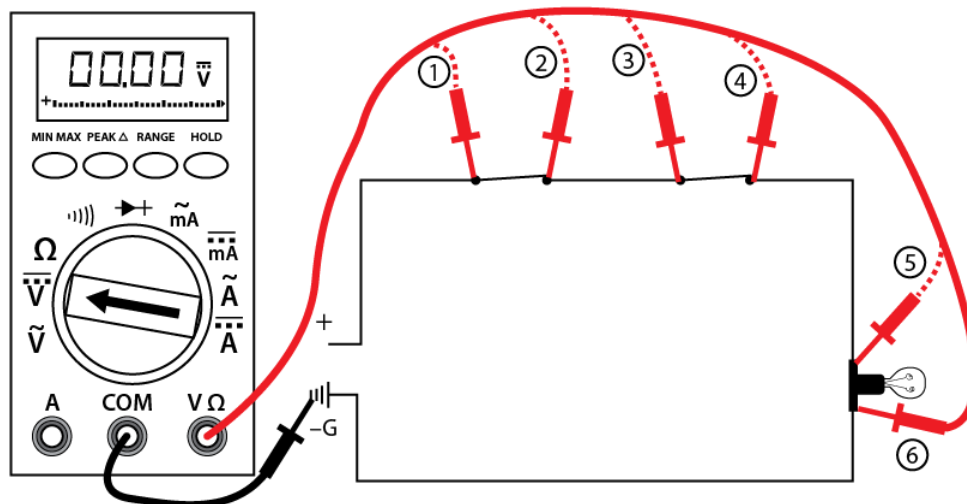


Figure 13.E.1: (CC BY-NC-SA; BC Industry Training Authority)

5. Refer to Figure 1. If you have a reading at 5 and zero at 6, what is the problem?
 1. The load has failed.
 2. The switch is open.
 3. The wiring has failed.
 4. The ground has failed.
6. What is always the first step when testing for resistance?
 1. Check the load.
 2. Remove the component.
 3. Turn the circuit power on.
 4. Turn the circuit power off.

7. What is the best test for checking if a heating element is faulty?
 1. Voltage test
 2. Continuity test
 3. Amperage test
 4. They cannot be tested.
8. What will increase when power sources are connected in series?
 1. Voltage
 2. Current
 3. Resistance
 4. Continuity

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14: Answer Key

Answer Key

Self-Test 1

1. c. Resistance
2. b. False
3. b. 2.2 V
4. b. False
5. c. A negative voltage would be read.
6. c. Using the one-hand technique
7. b. False
8. b. False
9. a. Series
10. c. After checking resistance
11. a. 0
12. c. 2125 Ω
13. c. Continuity test

Self-Test 2

1. c. 50 mA
2. a. Open
3. d. The load has failed.
4. c. The wiring has failed.
5. a. The load has failed.
6. d. Turn the circuit power off.
7. b. Continuity test
8. a. Voltage

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